

## DBMS: ASSIGNMENT 3

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### Aim:

SQL Queries- All types of Join, Sub-Query, and View:

- Write at least 10 SQL queries for a suitable database application using statements using joins, and subqueries.
- Write queries using the group by and having clause.

Enter password: \*\*\*\*

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 8

Server version: 8.0.45 MySQL Community Server - GPL

```
mysql> CREATE DATABASE Employee_Payroll_Analytics;
```

Query OK, 1 row affected (0.02 sec)

```
mysql> USE Employee_Payroll_Analytics;
```

Database changed

```
mysql> CREATE TABLE Division (
```

```
-> Div_ID INT PRIMARY KEY,
```

```
-> Div_Name VARCHAR(50)
```

```
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> CREATE TABLE Staff (
```

```
-> Staff_ID INT PRIMARY KEY,
```

```
-> Staff_Name VARCHAR(100),
```

```
-> Salary DECIMAL(10,2),
```

```
-> Div_ID INT,
```

```
-> FOREIGN KEY (Div_ID) REFERENCES Division(Div_ID)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE Performance_Record (
```

```
-> Record_ID INT PRIMARY KEY,
```

```
-> Staff_ID INT,
```

```
-> Performance_Score INT,
```

```
-> Incentive DECIMAL(10,2),
```

```
-> FOREIGN KEY (Staff_ID) REFERENCES Staff(Staff_ID)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO Division VALUES
```

```
-> (1, 'HR'),  
-> (2, 'Engineering'),  
-> (3, 'Finance'),  
-> (4, 'Operations'),  
-> (5, 'Cyber Security'),  
-> (6, 'Cloud'),  
-> (7, 'Support');
```

```
Query OK, 7 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Staff VALUES
```

```
-> (101, 'Adi Maitre', 50000, 1),  
-> (102, 'Kritika Chaudhari', 60000, 2),  
-> (103, 'Ritobrata Baksi', 55000, 3),  
-> (104, 'Sarang Kolekar', 70000, 2),  
-> (105, 'Soham Kulkari', 65000, 4),  
-> (106, 'Aryan Deshmukh', 48000, 5),  
-> (107, 'Neha Patil', 72000, 2),  
-> (108, 'Rohan Shah', 53000, 6);
```

```
Query OK, 8 rows affected (0.01 sec)
```

```
mysql> INSERT INTO Performance_Record VALUES
```

```
-> (1, 101, 4, 5000),  
-> (2, 102, 5, 8000),  
-> (3, 103, 3, 3000),  
-> (4, 104, 5, 9000),  
-> (5, 105, 4, 6000),  
-> (6, 106, 2, 2000),  
-> (7, 107, 5, 10000),  
-> (8, 108, 3, 3500);
```

```
Query OK, 8 rows affected (0.01 sec)
```

```
mysql> SELECT S.Staff_Name, D.Div_Name
```

```
-> FROM Staff S
```

```
-> INNER JOIN Division D ON S.Div_ID = D.Div_ID;
```

```
+-----+-----+  
| Staff_Name      | Div_Name      |  
+-----+-----+  
| Adi Maitre      | HR            |  
| Kritika Chaudhari | Engineering   |  
| Sarang Kolekar   | Engineering   |  
| Neha Patil       | Engineering   |  
| Ritobrata Baksi  | Finance       |
```

|                |                |  |
|----------------|----------------|--|
| Soham Kulkari  | Operations     |  |
| Aryan Deshmukh | Cyber Security |  |
| Rohan Shah     | Cloud          |  |
| +-----+-----+  |                |  |

```
mysql> SELECT Staff_Name, Salary
-> FROM Staff
-> WHERE Salary = (SELECT MAX(Salary) FROM Staff);
```

|               |          |
|---------------|----------|
| +-----+-----+ |          |
| Staff_Name    | Salary   |
| +-----+-----+ |          |
| Neha Patil    | 72000.00 |
| +-----+-----+ |          |

```
mysql> SELECT Staff_Name, Salary
-> FROM Staff
-> WHERE Salary > (SELECT AVG(Salary) FROM Staff);
```

|                   |          |
|-------------------|----------|
| +-----+-----+     |          |
| Staff_Name        | Salary   |
| +-----+-----+     |          |
| Kritika Chaudhari | 60000.00 |
| Sarang Kolekar    | 70000.00 |
| Soham Kulkari     | 65000.00 |
| Neha Patil        | 72000.00 |
| +-----+-----+     |          |

```
mysql> SELECT D.Div_Name, SUM(S.Salary) AS Total_Salary
-> FROM Staff S
-> JOIN Division D ON S.Div_ID = D.Div_ID
-> GROUP BY D.Div_Name;
```

|                |              |
|----------------|--------------|
| +-----+-----+  |              |
| Div_Name       | Total_Salary |
| +-----+-----+  |              |
| HR             | 50000.00     |
| Engineering    | 202000.00    |
| Finance        | 55000.00     |
| Operations     | 65000.00     |
| Cyber Security | 48000.00     |
| Cloud          | 53000.00     |
| +-----+-----+  |              |

```
mysql> SELECT D.Div_Name, AVG(S.Salary) AS Avg_Salary
-> FROM Staff S
-> JOIN Division D ON S.Div_ID = D.Div_ID
```

-> GROUP BY D.Div\_Name  
-> HAVING AVG(S.Salary) > 60000;

```
+-----+-----+
| Div_Name | Avg_Salary |
+-----+-----+
| Engineering | 67333.333333 |
| Operations | 65000.000000 |
+-----+-----+
```

mysql> CREATE VIEW Payroll\_Performance\_View AS  
-> SELECT S.Staff\_Name, D.Div\_Name, P.Performance\_Score, P.Incentive  
-> FROM Staff S  
-> JOIN Division D ON S.Div\_ID = D.Div\_ID  
-> JOIN Performance\_Record P ON S.Staff\_ID = P.Staff\_ID;  
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT \* FROM Payroll\_Performance\_View;

```
+-----+-----+-----+-----+
| Staff_Name | Div_Name | Performance_Score | Incentive |
+-----+-----+-----+-----+
| Adi Maitre | HR | 4 | 5000.00 |
| Kritika Chaudhari | Engineering | 5 | 8000.00 |
| Ritobrata Baksi | Finance | 3 | 3000.00 |
| Sarang Kolekar | Engineering | 5 | 9000.00 |
| Soham Kulkari | Operations | 4 | 6000.00 |
| Aryan Deshmukh | Cyber Security | 2 | 2000.00 |
| Neha Patil | Engineering | 5 | 10000.00 |
| Rohan Shah | Cloud | 3 | 3500.00 |
+-----+-----+-----+-----+
```